



Supplementary information to technical manual document reference 30/issue E.

The information in this user notice is provisional. It will, after checking, be incorporated in a future amendment to the manual.

Figure 5.8 Head
adjustment, oscilloscope
displays

Delete existing illustration. Insert: "For revised illustration see user notice number 30 001"

Section 5.5.5.5
Adjusting the
read/write heads

Delete the existing text. Insert: "For revised text see user notice number 30 001"

Revised text:

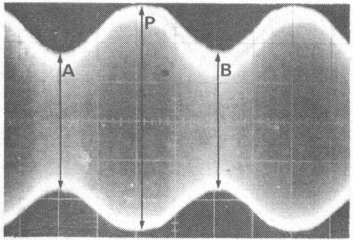
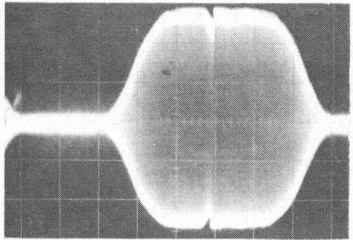
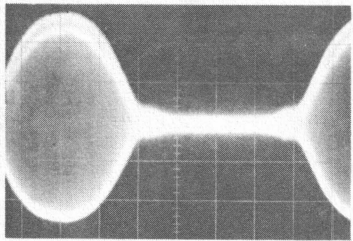
- (a) Remove power to the drive by interrupting the ac supply to the power supply. Remove interface connectors and fit a terminator to the interface socket.
- (b) For standard density drives only, fit head adaptor TD5083 between the head-lead plug and the head-lead socket of board J10.
- (c) Restore power and load the CE cartridge (part numbers TD277 and TD276 for 1100 and 2200 bpi respectively).
- (d) Allow the drive to run for 30 minutes for temperature stabilization
- (e) Position the head carriage at cylinder 100 (use cylinder 105 with a high-density drive).
- (f) Set up an oscilloscope as shown in table 5.6.

control	setting for standard-density drive	setting for high-density drive
timebase	10 ms/cm	5 ms/cm
deflexion sensitivity	10 mV/cm	1 V/cm
trigger	ext -ve	ext -ve

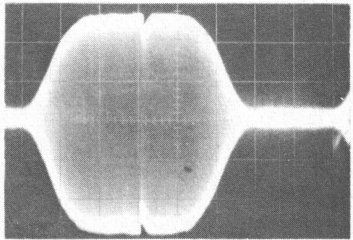
TABLE 5.6

HIGH DENSITY

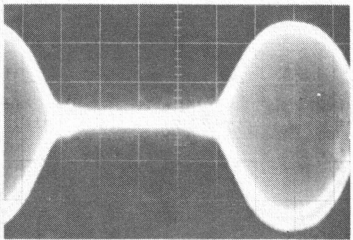
TOWARD
TRACK
104



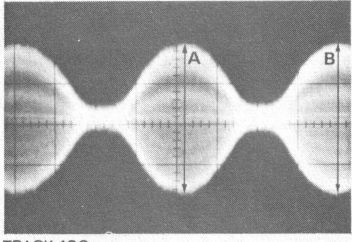
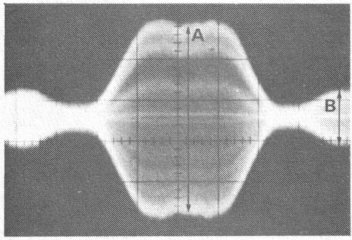
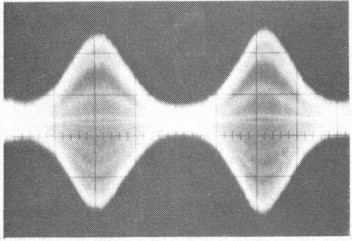
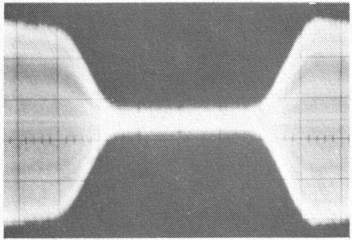
TRACK 105



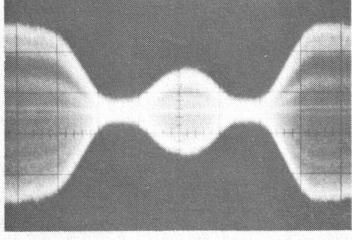
TOWARD
TRACK
106



TOWARD
TRACK
099



TRACK 100



TOWARD
TRACK
101

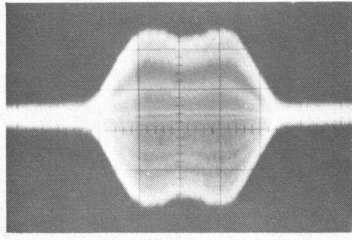
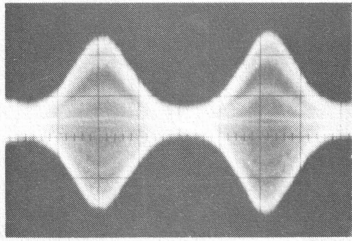


FIGURE 5.8 HEAD ADJUSTMENT, OSCILLOSCOPE DISPLAYS

- (g) Trigger an oscilloscope from the leading-edge of the index mark (this is the negative-going edge of the pulse at TP6 of the circuit board J9).
- (h) Monitor TP2 of J10 to check the alignment on both heads. With the interface connector removed the lower head is selected; the upper head is selected by grounding pin 5 of J10. Head alignment is checked by comparing the difference between successive lobes which, for a perfectly aligned head, corresponds with figure 5.8 (equal size of lobes).

Since all CE packs are slightly different from each other within tolerable limits, the extreme sensitivity of the head-alignment measuring technique will produce different results with different packs, and slight variations with the same pack. Actual differences in terms of head displacement are negligible. Owing to different head configurations, results for standard density drives will be different from those for high density ones. Thermal variations will also affect results. All these variations have been allowed for in the design and data interchangeability is assured provided the unit meets the following requirements.

High density drives are checked using the nulls of the oscilloscope trace. Resetting is unnecessary if nulls do not differ by more than 30% using the equation $\frac{A-B}{p} \times 100$.

If necessary, adjust until they are equal. See subsection (j).

Standard density drives are set using the peaks of the oscilloscope trace. Resetting is not necessary until successive peaks vary by more than 50% using the equation $\frac{A-B}{A+B} \times 100$.

However, owing to differences in the configurations between high and standard density heads, and the normal CE pack variations, the shape of the waveform alters when the peak amplitudes differ by more than 30% and the lobes may become indeterminate.

For this reason the following recommendation is made.

If the peak amplitude variation is seen to be within 30% then no adjustment should be made. If the lobes are indeterminate then the heads may be adjusted at the users discretion. This should only be carried out if there is some evidence of data incompatibility between the subject drive and other series 30 units.

It should be borne in mind that units newly delivered from Data Recording will have been set up by factory technicians using the CE packs which have been specially selected to be close to the centre line of the normal spread of such packs. It is therefore highly unlikely that new drives will require any adjustment to the head setting, though lobe discrepancies will probably be observable as indicated above.

- (j) If a head needs adjusting, slacken the three head clamp screws E, F and G and push heads against back of carriage. Retighten screws E and F to about 0,14 newton-metres (20 ounce-inches).
- (k) Using the head-adjusting tool (TD10289) move the selected head forward by screwing the tool against the head at points H or J (see figure 5.7). The head can be adjusted forwards only.
- (m) Heads are set using the erase windings for the standard density drive. The read/write winding is used for the high density drive. Either type is correctly adjusted when directly over the effective centre of the single eccentrically recorded track. Take care when adjusting heads; do not allow the tang on the end of the head support arm to disengage from the slot in which it is located (X in figure 5.7).
- (n) If the head is adjusted too far forward, unscrew the setscrew; switch to the load condition, remove the CE cartridge, loosen both clamps that secure the selected head, and manually push the head towards the rear of the drive. Do not push on the actual head pad or its mounting gimbals - use part of the rigid mounting hardware. Tighten the single-clamp screw to a torque of 0,18 newton-metres (26 ounce-inches) and repeat the adjusting procedure until the oscilloscope display is correct.
- (p) When the head is correctly adjusted, apply a torque of 0,56 newton-metres (80 ounce inches) to the double-clamp screw (G in figure 5.7). Tighten single-clamp screws (E and F) to the same torque.
- (q) Recheck head setting on oscilloscope.

Delete sections (f) to (j). Insert the following:

- (f) To check direction of trace, touch the X-plate lead on the +5 volt rail. If the trace moves to the right, the adjustment pattern will be the same as in figure 5.9. If the trace moves to the left the lissajous figure will be reversed left to right.

Section 5.5.5.6
Adjusting the
cylinder 000 stop